<u>CO2-AT1-SCENARIO BASED PROGRAMS</u>

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COURSE CODE: CSA0904

COURSE NAME: PROGRAMMING FOR JAVA

1.Suppose that a university system stores student data, where each student must be unique and can enroll in multiple courses.

(a) Enumerate three Collection interfaces suitable for this system.

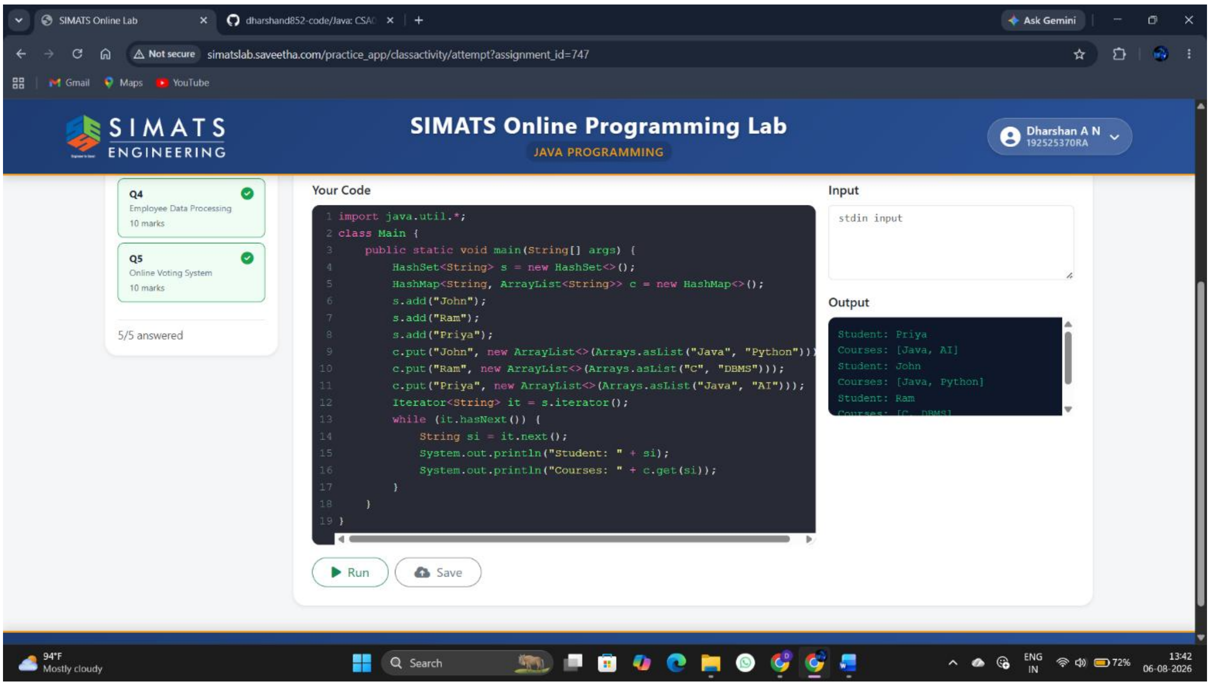
(b) Design a Collection structure (diagram/representation) to store students and their enrolled courses.

(c) Write a Java program using HashSet and HashMap with Generics to:

. Add students

. Map students to courses

. Display data using Iterator



2. Suppose that an online shopping system maintains a cart where products are added, removed, and displayed in order.

(a) List three Collection classes applicable for cart management.

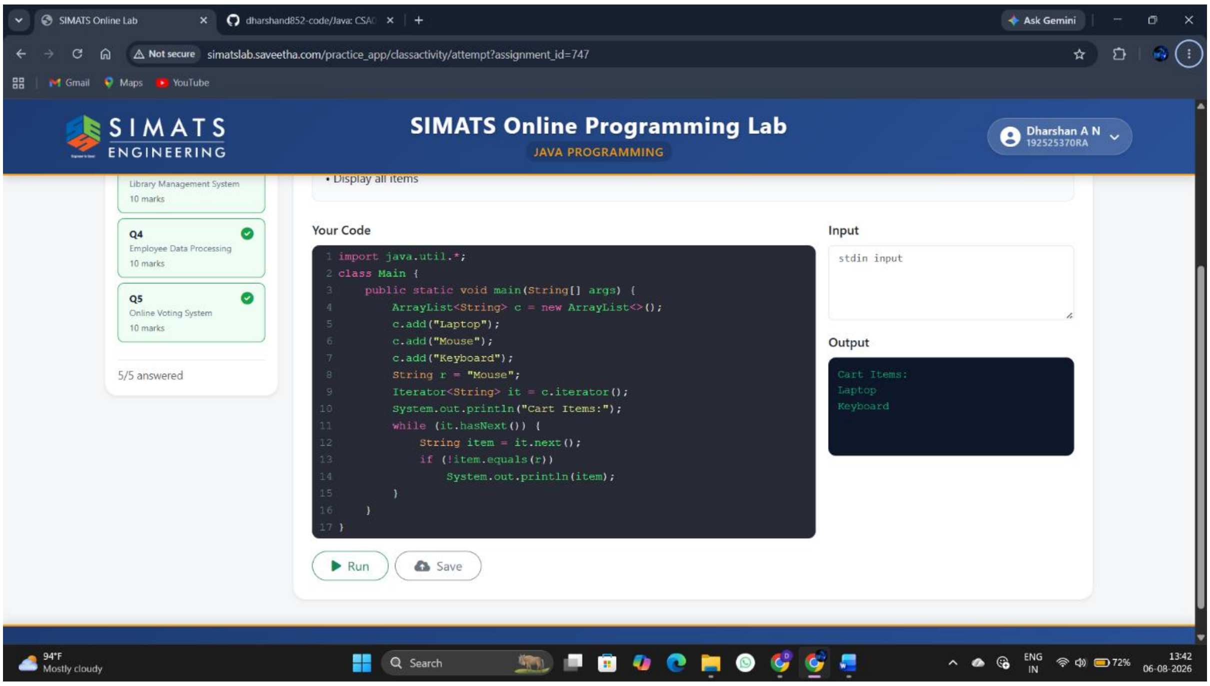
(b) Represent how cart items are stored using a List-based structure.

(c) Write a Java program using ArrayList and Iterator to:

. Add items

. Remove an item

. Display all items



3. Suppose that a library stores books with unique ISBN numbers and needs fast retrieval.

(a) Name three Map implementations in Java.

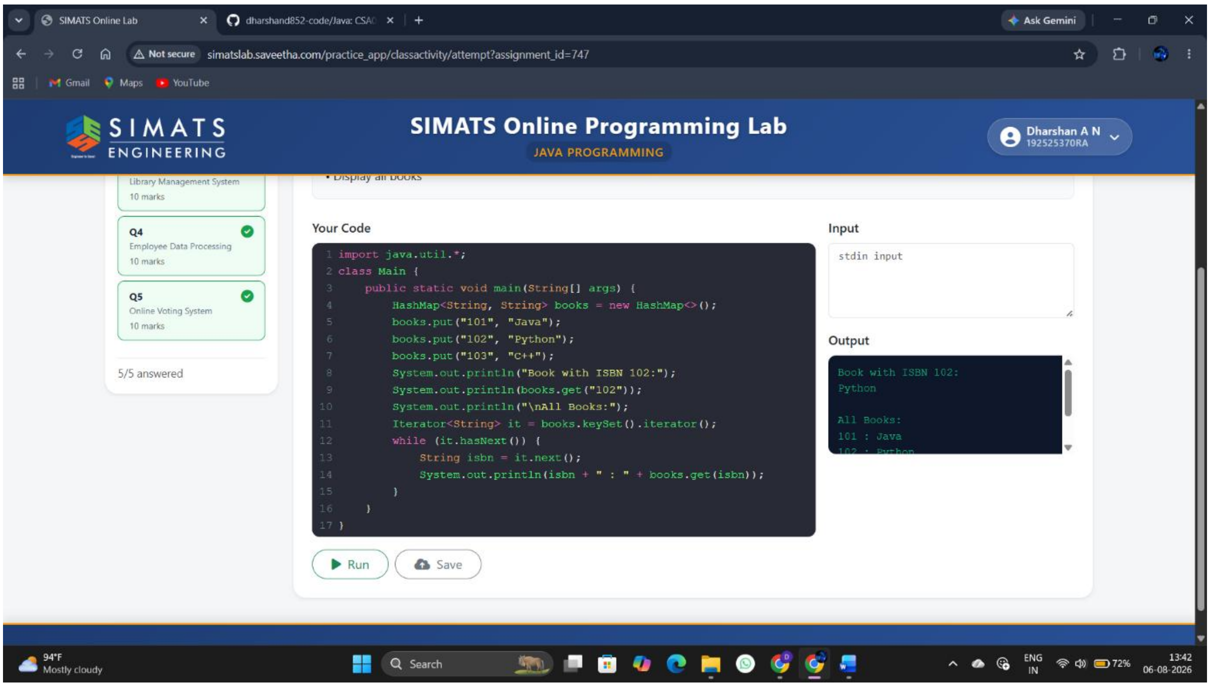
(b) Design a schema-like representation using Map for storing ISBN and book details.

(c) Write a Java program using HashMap to:

. Insert book details

. Retrieve a book

. Display all books



4. Suppose that a company processes employee records and filters employees based on salary.

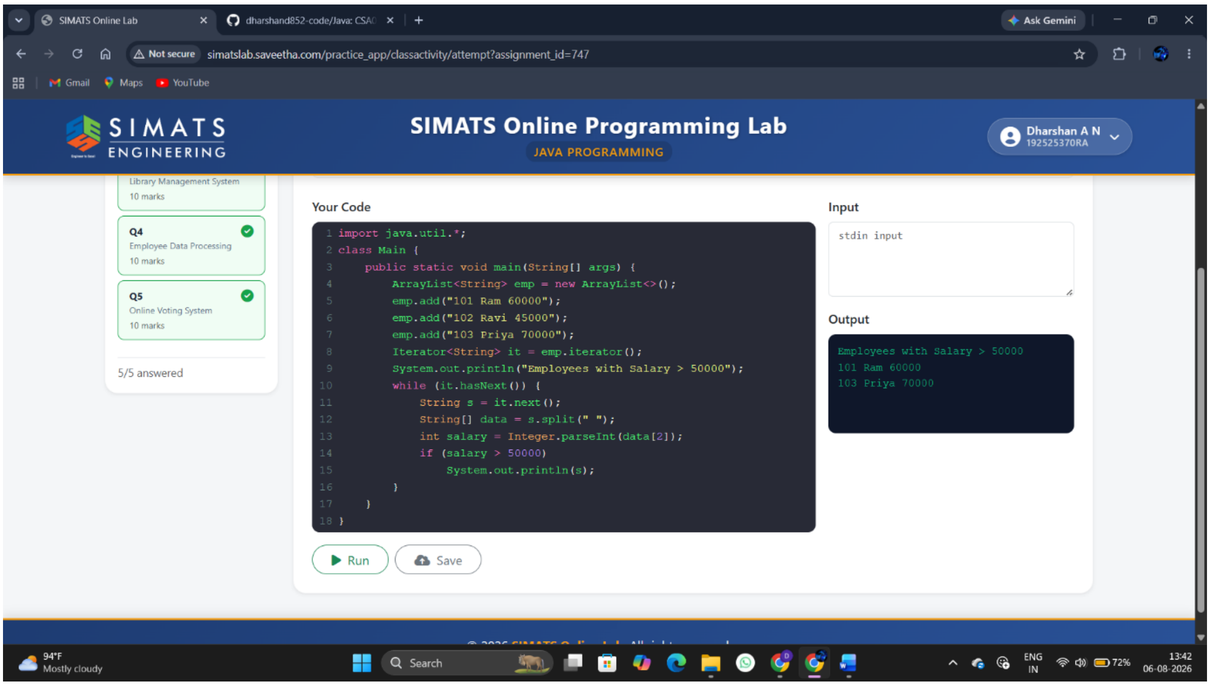
(a) List three advantages of Generics in Collections.

(b) Represent how employee data is stored using List with Generics.

(c) Write a Java program using Iterator to:

. Store employee objects

. Display employees with salary > 50,000



5. Suppose that an online voting system ensures one vote per user and counts votes efficiently.

(a) Identify three suitable Collection types for this system.

(b) Design a structure using Set and Map for voters and vote counting.

(c) Write a Java program using HashSet and HashMap to:

. Record votes

. Prevent duplicate voting . Display results